

Subject: Mathematics

Total Marks: 20

Time Allowed: 30 minutes

Passing Marks: 12

Instructions: Answer ALL questions. Show your working where required.

Q1. [2 marks]

What is the value of x in the equation: $2x + 6 = 14$?

Answer / Working:

Q2. [3 marks]

Factorise the following expression completely:

$$6x^2 + 9x$$

Answer / Working:

Q3. [4 marks]

Solve the simultaneous equations:

$$2x + y = 10$$

$$x - y = 2$$

Answer / Working:

Q4. [5 marks]

A right-angled triangle has legs of length 6 cm and 8 cm.

(a) Calculate the length of the hypotenuse. [2 marks]

(b) Calculate the area of the triangle. [1 mark]

(c) Calculate the perimeter of the triangle. [2 marks]

Answer / Working:

Q5. [6 marks]

Given the quadratic equation: $x^2 - 5x + 6 = 0$

- (a) Factorise the equation. [2 marks]
(b) Find both roots of the equation. [2 marks]
(c) Verify answers by substituting back. [2 marks]

Answer / Working:
